

Orthopaedic Conditions of the Hip

The Big Picture

Three big topics: **AVN (osteonecrosis)** of the femoral head, **hip osteoarthritis (OA)**, and **total hip replacement (THR)** — the "operation of the century." Common thread: dying or worn-out femoral head → pain + lost function → staged treatment from drugs to joint replacement.

Avascular Necrosis (Hip Osteonecrosis)

- **What** — reduced blood flow to femoral head → bone death → collapse.
- **Who** — male > female, **age 35–50**; **bilateral 80%** of the time. ~10% of all hip arthroplasties.
- **Causes** — direct: trauma (injures **medial femoral circumflex**), irradiation, leukaemia/lymphoma, **Caisson/decompression (the bends)**, Gaucher's, **sickle cell**. Indirect: **alcohol**, **steroids**, **SLE**, hypercoagulable, transplant, HIV/protease inhibitors, idiopathic.
- **Pathophysiology** — intravascular coagulation → venous thrombosis → ↑ intraosseous pressure → ↓ flow → AVN → **chondral fracture & collapse**.
- **Presentation** — insidious **anterior hip pain**, worse on stairs/impact; exam normal early, later like OA (↓ internal rotation).
- **Imaging** — X-ray (AP + frog-lateral); **MRI = 99% sens/spec**, "double density," T1 dark band, T2 bright (marrow oedema = predicts progression).

Steinberg Classification (modified Ficat)

- 0 normal | I normal X-ray, abnormal MRI | II cyst/sclerosis | **III crescent sign (subchondral collapse)** | IV head flattening | V joint narrowing | VI advanced degeneration.

AVN Treatment (stage-driven)

- **Bisphosphonates** — pre-collapse (Ficat 0–II).
- **Core decompression** ± bone graft — early, pre-collapse, reversible cause; relieves pressure + triggers angiogenesis.
- **Rotational osteotomy** — small lesions (<15%), rotate off weight-bearing zone.
- **Curettage + graft** — lightbulb (through neck-head junction) or trapdoor (through articular surface).
- **Vascularised free-fibula transfer** — young patient, pre/post collapse; donor-site morbidity.
- **Total hip replacement** — crescent sign/collapse, irreversible cause (chronic steroids), >40 with large lesions. Young AVN patients get **more liner wear/osteolysis** than older OA patients.

Hip Osteoarthritis

- **What** — degenerative loss of femoral head + acetabular cartilage.
- **Risk** — modifiable (trauma, muscle weakness, heavy work, high-impact sport); non-modifiable (**female > male**, age, genetics, dysplasia, SCFE, Perthes).
- **Pathology** — ↑ cartilage water, ↓ proteoglycans; **MMP/TIMP imbalance**; cytokines IL-1, IL-6, TNF-α. Late: subchondral cysts + sclerosis.
- **Presentation** — function-limiting **groin pain**, ↓ walking distance, night/rest pain, stiffness, **limited internal rotation**, flexion contracture.
- **Imaging** — standing AP pelvis: **joint space narrowing**, **osteophytes**, **subchondral sclerosis + cysts**.

Tonnis Classification

- **0 normal** | 1 sclerosis + slight narrowing + lipping | 2 large cysts + severe narrowing + ↓ sphericity | 3 severe head deformity.

OA Treatment

- **Non-op (first line)** — NSAIDs/tramadol, **walking stick in opposite hand**, weight loss/exercise/PT, steroid injection (Dx + Rx).
- **Operative** — arthroscopic debridement (labral tears), **periacetabular osteotomy** (young dysplasia, delays THR), hip resurfacing (young active males), **THA** (end-stage, preferred >50).

Total Hip Replacement (THR)

- **The 4 demands** — implant durable, **low-friction** articulation, firmly fixed to bone, inert (no tissue reaction).
- **History** — Gluck ivory (1891), Austin Moore metal hemiarthroplasty (1940), **Sir John Charnley low-friction arthroplasty (1960s)** = small metal head + polyethylene cup + acrylic cement. **~80% survival at 25 years.**
- **Indications** — pain, lost movement, disability + radiographic joint destruction.
- **Contraindication** — **sepsis (overt or latent)** is the chief one — infected arthroplasty = disaster. <60yo only if no alternative.
- **Biomechanics depend on** — design, bearing surface/lubrication, fixation.
- **Fixation** — press-fit (biologic, hoop stresses) vs cemented (grout; preferred for irradiated bone; cobalt-chrome commonest).
- **Bearing surfaces** — metal-on-polyethylene, metal-on-metal, ceramic-on-ceramic, ceramic-on-poly, titanium-on-poly.
- **Complications** — **Acute:** infection, dislocation, sciatic nerve palsy, vascular injury, periprosthetic fracture, leg-length discrepancy. **Chronic: aseptic loosening**, iliopsoas impingement, heterotopic ossification, infection.

Walk-Away Points

- AVN: bilateral 80%, alcohol/steroid/sickle cell; MRI 99%; **crescent sign = collapse → THR.**
- Stage AVN treatment: pre-collapse = core decompression; collapsed/old = THR.
- Hip OA: groin pain + lost internal rotation; **Tonnis** grading; non-op first, THA >50.
- THR: **Charnley low-friction**, ~80% at 25yr; **sepsis = chief contraindication**; aseptic loosening = main long-term failure.

Hip Orthopaedic Conditions — Revision Table	
AVN of hip	Detail
Definition	↓ blood flow → femoral head death → collapse
Demographics	Male>female, 35–50yo, bilateral 80%
Direct causes	Trauma (medial femoral circumflex), irradiation, sickle cell, Caisson, Gaucher
Indirect causes	Alcohol, steroids, SLE , hypercoagulable, HIV
Pathophysiology	Coagulation → venous thrombosis → ↑ intraosseous pressure → collapse
Symptoms	Insidious anterior hip pain , worse stairs/impact
Imaging	MRI 99% sens/spec ; T1 dark band, T2 bright (oedema)
Steinberg/Ficat stage	Finding
0–I	Normal X-ray; abnormal MRI
II	Cyst / sclerosis
III	Crescent sign (subchondral collapse)
IV	Head flattening
V	Joint narrowing
VI	Advanced degeneration
AVN treatment	When
Bisphosphonates	Pre-collapse (0–II)
Core decompression	Early, pre-collapse, reversible
Rotational osteotomy	Small lesion <15%
Free-fibula transfer	Young, pre/post collapse
THR	Crescent/collapse, >40, irreversible cause
Hip OA	Detail
Risk	Female>male, trauma, dysplasia, SCFE, Perthes
Pathology	↓ proteoglycans, MMP/TIMP imbalance, IL-1/6/TNF-α
Symptoms	Groin pain , ↓ internal rotation, stiffness
X-ray	Narrowing, osteophytes, sclerosis, cysts
Tonnis grade	Finding
0	Normal
1	Sclerosis + slight narrowing
2	Large cysts + severe narrowing
3	Severe head deformity
OA treatment	
First line	NSAID/tramadol, stick in opposite hand , weight loss, PT, steroid injection
Operative	PAO (young dysplasia), resurfacing (young male), THA (>50, end-stage)
Total Hip Replacement	Detail
Charnley	Low-friction: metal head + poly cup + cement
Survival	~80% at 25 years
Chief contraindication	Sepsis (overt/latent)
Fixation	Press-fit (biologic) vs cemented (grout)

Total Hip Replacement	Detail
Bearing	Metal/ceramic/titanium on polyethylene
Acute complications	Infection, dislocation, sciatic palsy, periprosthetic fracture, LLD
Chronic complications	Aseptic loosening , HO, impingement

Bold = high-yield.